ARJUNA JEE 2027

Lecture -11 | Mathematics

SEQUENCE & SERIES

By- Sachin Jakhar Sir



Topics *to be covered*

A **Telescopic Method of Difference**

B **Inequality**

C **Question Practice**

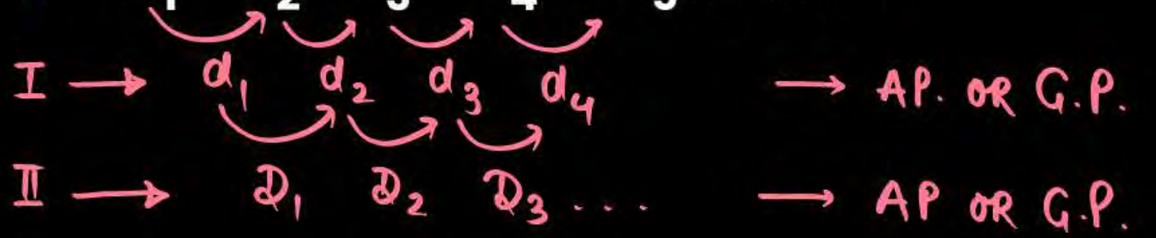
D





SHORTCUT METHOD TO FIND T_n

Sequence : a_1 a_2 a_3 a_4 a_5 ... .



If Ist – difference is in A.P., then $T_n = an^2+bn+c$

If IInd – difference is in A.P., then $T_n = an^3+bn^2+cn+d$

If Ist – difference is in G.P., then $T_n = a(cR)^n+bn+c$

If IInd – difference is in G.P., then $T_n = a(cR)^n+bn^2+cn+d$

Question Practice

Challenger Question

If $\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\cdots\infty=\frac{\pi^2}{6}$, then choose correct option -

A $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\cdots\infty=\frac{\pi^2}{8}$

$\left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\ldots\infty\right)-\left(\frac{1}{2^2}+\frac{1}{4^2}+\frac{1}{6^2}+\ldots\infty\right)$

$\frac{\pi^2}{6}-\frac{1}{2^2}\left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\ldots\infty\right)$

$\frac{\pi^2}{6}-\frac{1}{4}\times\frac{\pi^2}{6}=\frac{\pi^2}{6}\left(1-\frac{1}{4}\right)$

$=\frac{\pi^2}{6}\times\frac{3}{4}=\frac{\pi^2}{8}$

B $\frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+\cdots\infty=\frac{\pi^2}{12}$

$\left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\ldots\infty\right)$

$\left(-2\left(\frac{1}{2^2}+\frac{1}{4^2}+\frac{1}{6^2}+\ldots\infty\right)\right.$

$\frac{\pi^2}{6}-2\times\frac{\pi^2}{24}=\frac{\pi^2}{12}$

C $\frac{1}{4^2}+\frac{1}{8^2}+\frac{1}{12^2}+\cdots\infty=\frac{\pi^2}{96}$

D $\frac{1}{2^2}+\frac{1}{4^2}+\frac{1}{6^2}+\cdots\infty=\frac{\pi^2}{24}$

Question

Find the sum of the following series to n terms :

$5 + 7 + 13 + 31 + 85 + \ldots\ldots\ldots$

2 → 6 → 18 → 54 ---- → G.P. (CR = 3)

Ans: $\boxed{T_n = 3^{n-1} + 4}$

$$T_n = a(3)^n + bn + c$$

$$T_1 = 3a + b + c = 5$$
$$T_2 = 9a + 2b + c = 7$$
$$T_3 = 27a + 3b + c = 13$$

OR

$$\# \quad \boxed{T_n = \frac{1}{3} \times 3^n + 4}$$

$$S_n = \sum T_n = \sum\left(\frac{1}{3}3^n + 4\right)$$

$$= \frac{1}{3}\left(\underbrace{\sum_{n=1}^{n} 3^n}_{GP} + \underbrace{\sum_{n=1}^{n} 4}_{4n}\right)$$

Question [JEE Mains-2023]

90%

Let a_n be the nth term of the series $5 + 8 + 14 + 23 + 35 + 50 + \ldots\ldots$ and $S_n = \sum_{k=1}^{n} a_k$. Then $S_{30} - a_{40}$ equal to

3 6 9 12 15 ---- → A.P.

$T_{40} = a(40)^2 + b(40) + c$

$T_n = an^2 + bn + c$

$a = \frac{3}{2}, b = -\frac{3}{2}, c = 5$

$$S_{30} = \sum_{n=1}^{30} \left(\frac{3}{2}n^2 - \frac{3}{2}n + 5\right)$$

$$= \frac{3}{2}\sum_{n=1}^{30} n^2 - \frac{3}{2}\sum_{n=1}^{30} n + \sum_{r=1}^{30} 5$$

A 11310

B 11280

C 11290

D 11260

Question [JEE Mains-2023]

Let a_n be the nth term of the series $5 + 8 + 14 + 23 + 35 + 50 + \ldots\ldots$ and $S_n = \sum_{k=1}^{n} a_k$. Then $S_{30} - a_{40}$ equal to

3 6 9 12 15 ----→ A.P.

$a_{40} = ?$

$T_n = \frac{3}{2}n^2 - \frac{3}{2}n + 5$

$r = 40$

$T_n = an^2 + bn + c$.

$T_1 = a + b + c = 5$

$T_2 = 4a + 2b + c = 8$

$T_3 = 9a + 3b + c = 14$

$\Rightarrow 3a + b = 3$

$\Rightarrow 5a + b = 6$

$2a = 3$

$a = \frac{3}{2}$

$b = -\frac{3}{2}$

$c = 5$

$S_{30} = \sum_{r=1}^{30} T_r = \frac{3}{2}\sum_{r=1}^{30} n^2 - \frac{3}{2}\sum_{n=1}^{30} n + \sum_{r=1}^{30} 5$

(A) 11310

(B) 11280

(C) 11290

(D) 11260



LAST CLASS RECAP

Telescopic Method of Difference: (TMOD)

Type-01: *Terms of AP is in multiplication in Numerator or Denominator:*

Ex. : $S = \overset{+3}{\overbrace{1}} \times 4 \times 7 + \overset{+3}{\overbrace{2}} \times 5 \times 8 + \overset{+3}{\overbrace{3}} \times 6 \times 9 + \ldots\ldots$ upto n-terms.

$$\Rightarrow \quad T_n = n(n+3)(n+6)$$

$$\Rightarrow \quad S_n = \sum_{n=1}^{n} \frac{n(n+3)(n+6)}{12} \left(\overbrace{(n+9)-(n-3)}^{12} \right) = \sum_{n=1}^{n} \left(\frac{n(n+3)(n+6)(n+9)}{12} - \frac{(n-3)n(n+3)(n+6)}{12} \right)$$

Question

[Ans. : B]

Find the sum of the series $1 \cdot 2 \cdot 3 \cdot 4 + 2 \cdot 3 \cdot 4 \cdot 5 + 3 \cdot 4 \cdot 5 \cdot 6 + \ldots\ldots$ up to n terms

A $\frac{1}{4} n(n+1)(n+2)(n+3)$

B $\frac{1}{5} n(n+1)(n+2)(n+3)(n+4)$

C $\frac{1}{5} (n-1)n(n+1)(n-2)(n+3)$

D $\frac{1}{6} n(n+1)(n+2)(n+3)(n+4)(n+5)$

QIBY – (A)

TYPE-01: CONTINUOUS TERMS OF A.P. ARE IN MULTIPLICATION IN NUMERATOR OR DENOMINATOR

Question

Solve : $S = \frac{1}{1\times2\times3} + \frac{1}{2\times3\times4} + \frac{1}{3\times4\times5} + \cdots$ upto 10 Terms.

$$T_n = \frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left(\frac{(n+2)-(n)}{n(n+1)(n+2)}\right) = \frac{1}{2}\left(\frac{\cancel{n+2}}{n(n+1)\cancel{(n+2)}} - \frac{n}{n(n+1)(n+2)}\right)$$

$$T_n = \frac{1}{2}\left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right)$$

Pehle aur aakhri ka difference Numerator में बनाओ

$$T_n = \frac{1}{2}\left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right)$$

$$T_1 = \frac{1}{2}\left(\frac{1}{1\times 2} - \frac{1}{2\times 3}\right)$$

$$T_2 = \frac{1}{2}\left(\frac{1}{2\times 3} - \frac{1}{3\times 4}\right)$$

$$T_3 = \frac{1}{2}\left(\frac{1}{3\times 4} - \frac{1}{4\times 5}\right)$$

$$\vdots$$

$$T_{10} = \frac{1}{2}\left(\frac{1}{10\times 11} - \frac{1}{11\times 12}\right)$$

$$S_{10} = \frac{1}{2}\left(\frac{1}{2} - \frac{1}{11\times 12}\right) \text{ Ans.}$$

Ques:-

$$S = \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{1}{1\times 2} + \frac{1}{2\times 3} + \frac{1}{3\times 4} + \frac{1}{4\times 5} + \cdots\cdots \infty$$

$$\sum \frac{1}{n(n+1)} = \frac{(n+1)-n}{n(n+1)} = \sum_{n=1}^{\infty}\left(\frac{1}{n} - \frac{1}{n+1}\right)$$

Ques

$$S = \sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)}$$

$$T_1 = \frac{1}{1} - \frac{1}{2}$$

$$T_2 = \frac{1}{2} - \frac{1}{3}$$

$$T_3 = \frac{1}{3} - \frac{1}{4}$$

$$T_4 = \frac{1}{4} - \frac{1}{5}$$

$$\vdots$$

$$T_n = \frac{1}{n} - \frac{1}{n+1}$$

$$1 = S_\infty = 1 - \frac{1}{\infty}$$

$n \to \infty$, 0

$$S_n = 1 - \frac{1}{n+1}$$

Ques. $\xi = \sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)} = \frac{1}{2}$

$$\frac{1}{2}\frac{\overbrace{(2n+1)-(2n-1)}^{2}}{(2n-1)(2n+1)} = \frac{1}{2}\left(\frac{1}{2n-1} - \frac{1}{2n+1}\right)$$

$$T_1 = \frac{1}{2}\left(\frac{1}{1} - \frac{1}{3}\right)$$

$$= \frac{1}{2}\left(\frac{1}{3} - \frac{1}{5}\right)$$

$$= \frac{1}{2}\left(\frac{1}{5} - \frac{1}{7}\right)$$

$$\vdots$$

$$\infty$$


When D^r contains
Terms in multip.
T.M.O.D.

Acidic Question

$S = \frac{1}{1\times3} + \frac{2}{1\times3\times5} + \frac{3}{1\times3\times5\times7} + \cdots$ upto n-terms and also find sum of infinite terms.

$\frac{2}{15}$ $\frac{1}{35}$ $\frac{1}{3}$

$$T_n = \frac{n}{1\times3\times5\times7\times\ldots(2n+1)}$$

$$= \frac{1}{2}\left(\frac{(2n+1)-1}{1\times3\times5\times7\times\ldots(2n+1)}\right)$$

$$T_n = \frac{1}{2}\left(\frac{2n+1}{1\times3\times5\times\ldots(2n+1)} - \frac{1}{1\times3\times5\times\ldots(2n+1)}\right)$$

$$T_n = \frac{1}{2}\left(\frac{2n+1}{1\times3\times5\times\ldots(2n+1)} - \frac{1}{1\times3\times5\times\ldots(2n+1)}\right)$$

$$T_n = \frac{1}{2}\left(\frac{1}{1\times3\times5\times\ldots(2n-1)} - \frac{1}{1\times3\times5\times\ldots(2n+1)}\right)$$

$$T_1 = \frac{1}{2}\left(\frac{1}{1} - \frac{1}{1\times3}\right)$$

$$T_2 = \frac{1}{2}\left(\frac{1}{1\times3} - \frac{1}{1\times3\times5}\right)$$

$$T_3 = \frac{1}{2}\left(\frac{1}{1\times3\times5} - \frac{1}{1\times3\times5\times7}\right)$$

$$\vdots$$

$$T_n = \frac{1}{2}\left(\frac{1}{1\times3\times5\times\ldots(2n-1)} - \frac{1}{1\times3\times5\times\ldots(2n+1)}\right)$$

add

$$S_n = \frac{1}{2}\left(1 - \frac{1}{1\times3\times5\times\ldots(2n+1)}\right)$$

$n\to\infty$

$$S_\infty = \frac{1}{2}\left(1 - \frac{1}{\infty}\right)$$

$\frac{1}{\infty} \to 0$

$$S_\infty = \frac{1}{2}$$

Question [JEE Mains-2021]

Let $a_1, a_2, \ldots, a_{21}$ be an AP such that $\sum_{n=1}^{20} \frac{1}{a_n a_{n+1}} = \frac{4}{9}$. If the sum of this AP is 189, then $a_6 \cdot a_{16}$ is equal to

$$\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \frac{1}{a_3 a_4} + \ldots\ldots + \frac{1}{a_{20} a_{21}}$$

Hew.

A 57

B 72

C 48

D 36

TYPE-02: BHEJA – FRY!!
USE OF FACTORISATION

FACTORISATION

(i) $r^4 + r^2 + 1 = (r^2+r+1)(r^2-r+1)$

(ii) $r^4 + 2r^2 + 9$

(iii) $4r^4 + 1$

$r^4 + 9 + 2r^2$

$(r^2)^2 + (3)^2 + 2r^2(3) - 2r^2(3) + 2r^2$

$-6r^2 + 2r^2$

a b

$(r^2+3)^2 - (2r)^2 = (r^2+3+2r)(r^2+3-2r)$

(iii) $4x^4 + 1$

$(2x^2)^2 + (1)^2 + 2(2x^2)(1) - 2(2x^2)(1)$

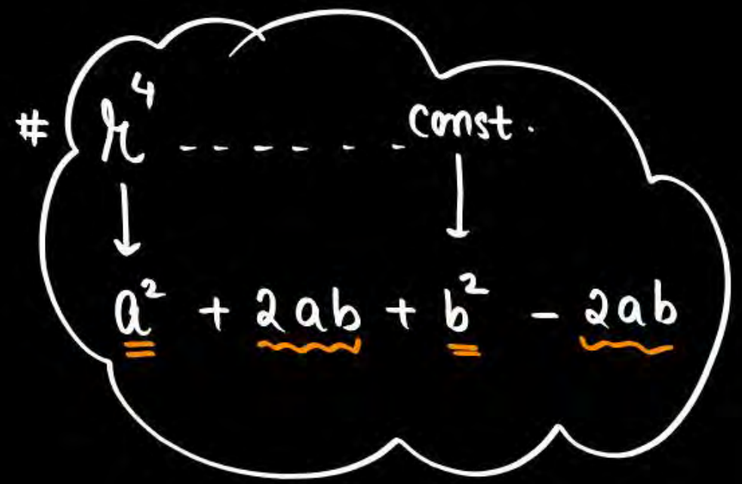
a, b

$(2x^2+1)^2 - 4x^2$

$(2x^2+1)^2 - (2x)^2 = (2x^2+1+2x)(2x^2+1-2x)$

420

$x^4 + x^2 + 1 = (x^2+x+1)(x^2-x+1)$



$x^4 + x^2 + 1$

$x^4 + 1 + x^2$

$(x^2)^2 + (1)^2 + 2x^2(1) - 2x^2(1) + x^2$

a b

$(x^2+1)^2 - x^2 = (x^2+1+x)(x^2+1-x)$

Question

If $\sum_{r=1}^{\infty} \frac{r}{r^4+r^2+1} = \frac{m}{n}$ where m & n are relatively prime, then

$m + n =$ _____

$$\frac{r}{r^4+r^2+1} = \frac{r}{(r^2-r+1)(r^2+r+1)} = \frac{1}{2}\left(\frac{(r^2+r+1)-(r^2-r+1)}{(r^2-r+1)(r^2+r+1)}\right)$$

$$T_r = \frac{1}{2}\left(\frac{1}{r^2-r+1} - \frac{1}{r^2+r+1}\right)$$

$$T_k = \frac{1}{2}\left(\frac{1}{k^2-k+1} - \frac{1}{k^2+k+1}\right)$$

$$T_1 = \frac{1}{2}\left(\frac{1}{1} - \frac{1}{3}\right)$$

$$T_2 = \frac{1}{2}\left(\frac{1}{3} - \frac{1}{7}\right)$$

$$T_3 = \frac{1}{2}\left(\frac{1}{7} - \frac{1}{13}\right)$$

$$\vdots$$

$$T_n = \frac{1}{2}\left(\frac{1}{n^2-n+1} - \frac{1}{n^2+n+1}\right)$$

$$S_n = \frac{1}{2}\left(1 - \frac{1}{n^2+n+1}\right) \xrightarrow{n\to\infty} S_\infty = \frac{1}{2}(1-0) = \frac{1}{2}$$

Question

Find $S = \sum_{r=1}^{\infty} \frac{r}{r^4+2r^2+9}$

HW.

$$\frac{r}{r^4+2r^2+9} = \frac{r}{(r^2+3+2r)(r^2+3-2r)} = \frac{1}{4}\left(\frac{(r^2+3+2r)-(r^2+3-2r)}{(r^2+3+2r)(r^2+3-2r)}\right)$$

Question [JEE Mains-2025]

If the sum of the first 10 terms of the series $\frac{4.1}{1+4.1^4}+\frac{4.2}{1+4.2^4}+\frac{4.3}{1+4.3^4}+\cdots$ is $\frac{m}{n}$, where $\gcd(m,n)=1$, then $m+n$ is equal _____.



$$T_n = \frac{4n}{1+4n^4}$$

Question (Involvement of Exponent

$5^{n+1} = 5^n \cdot 5^1$

Find the sum of series, $S = \frac{9}{1.2.5^2} + \frac{13}{2.3.5^3} + \frac{17}{3.4.5^4} + \ldots\ldots\infty$

$$T_n = \frac{(4n+5)}{n(n+1)\cdot 5^{n+1}} = \frac{(4n+5)}{n(5n+5)\cdot 5^n} = \frac{(5n+5)-n}{n(5n+5)\cdot 5^n}$$

$5^{n+1} = 5^n \cdot 5^1$

$$= \frac{5n+5}{n(5n+5)\cdot 5^n} - \frac{n}{n(5n+5)\cdot 5^n}$$

$$T_n = \frac{1}{n\cdot 5^n} - \frac{1}{(n+1)5^{n+1}}$$

$$T_n = \frac{1}{n \cdot 5^n} - \frac{1}{(n+1)5^{n+1}}$$

$$T_1 = \frac{1}{1 \times 5^1} - \frac{1}{2 \times 5^2}$$

$$T_2 = \frac{1}{2 \times 5^2} - \frac{1}{3 \times 5^3}$$

$$T_3 = \frac{1}{3 \times 5^3} - \frac{1}{4 \times 5^4}$$

$$\vdots$$

$$\infty$$

$$S_\infty = \frac{1}{5}$$ Ans.


INEQUALITY

3, 7

$AM = \frac{3+7}{2} = 5$

$GM = (3 \times 7)^{\frac{1}{2}} = \sqrt{21}$

$HM = \frac{2}{\frac{1}{3}+\frac{1}{7}} = \frac{2 \times 21}{10} = \frac{21}{5} = 4.2$

$AM > GM > HM$

3, 3

$AM = 3$

$GM = 3$

$HM = 3$

$AM \geq GM \geq HM$

Root Mean Square.

R.M.S.

$a_1 \quad a_2 \quad a_3$

Sq. $\rightarrow$ $a_1^2 \quad a_2^2 \quad a_3^2$

Mean $\rightarrow$ $\dfrac{a_1^2 + a_2^2 + a_3^2}{3}$

Root $\rightarrow$ $RMS = \sqrt{\dfrac{a_1^2 + a_2^2 + a_3^2}{3}}$

INEQUALITIES

For n positive numbers $(a_1, a_2, \ldots. a_n)$:

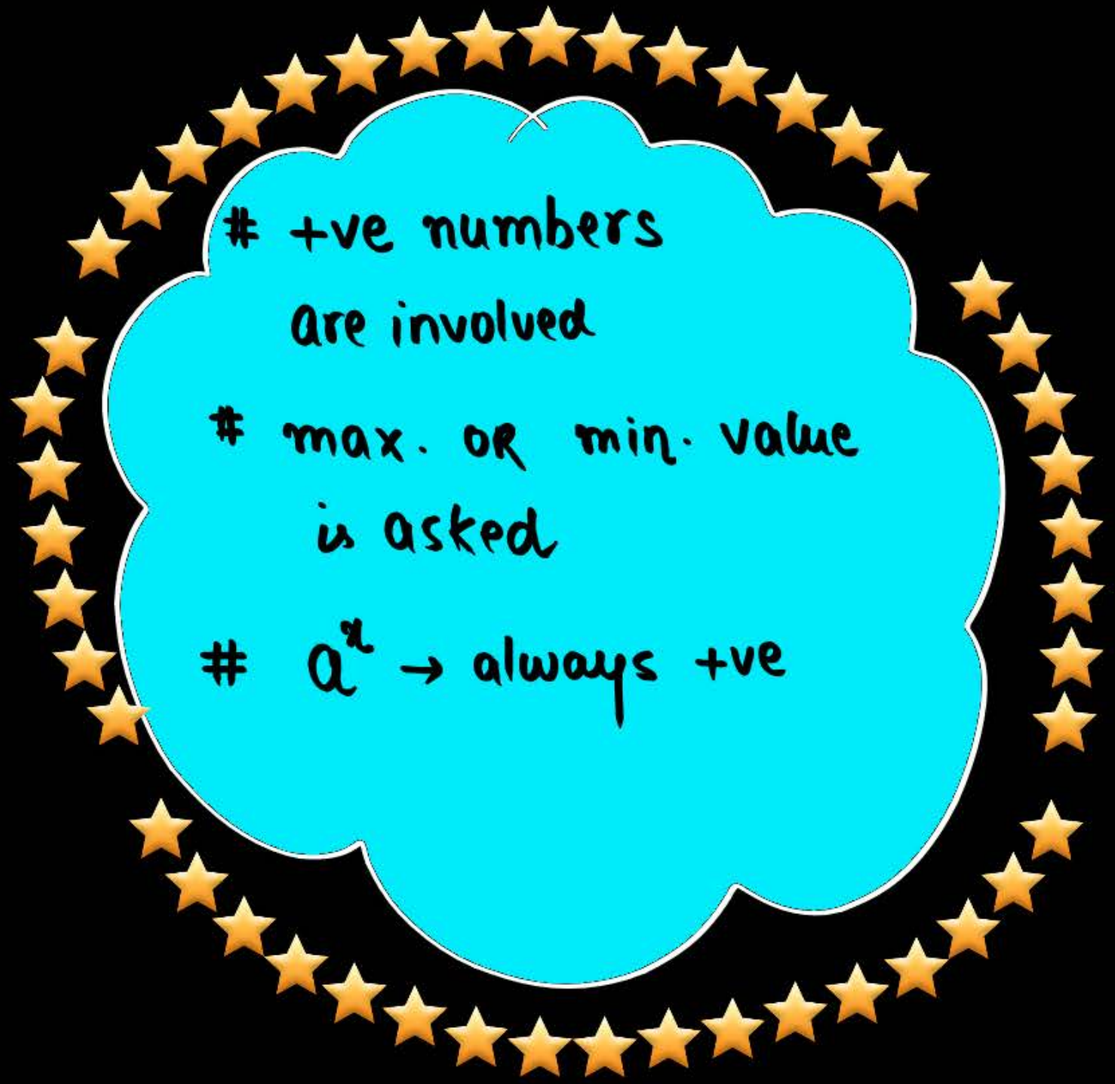
R.M.S. ≥ A.M. ≥ G.M. ≥ H.M.

$$\Rightarrow \sqrt{\frac{a_1^2+a_2^2+\ldots.+a_n^2}{n}} \geq \frac{a_1+a_2+\ldots+a_n}{n} \geq (a_1 a_2 a_3 \ldots. a_n)^{1/n} \geq \frac{n}{\frac{1}{a_1}+\frac{1}{a_2}+\ldots+\frac{1}{a_n}}$$

Equality holds when number are equal $(a_1 = a_2 = a_3 = \ldots.. = a_n)$

$\downarrow$

AM = GM = HM


+ve numbers are involved
max. or min. value is asked
a^2 → always +ve

Question

If $x > 0$, then minimum value $x + \frac{1}{x}$ is ____2____ at x = ____1____.

$x > 0 \Rightarrow x \to$ (+ve)

$x, \frac{1}{x} \longrightarrow AM \geq GM$

$\frac{x + \frac{1}{x}}{2} \geq \left(x \times \frac{1}{x}\right)^{\frac{1}{2}}$

$x + \frac{1}{x} \geq 2$

$x + \frac{1}{x}\Big|_{\min.} = 2$

$x = \frac{1}{x}$

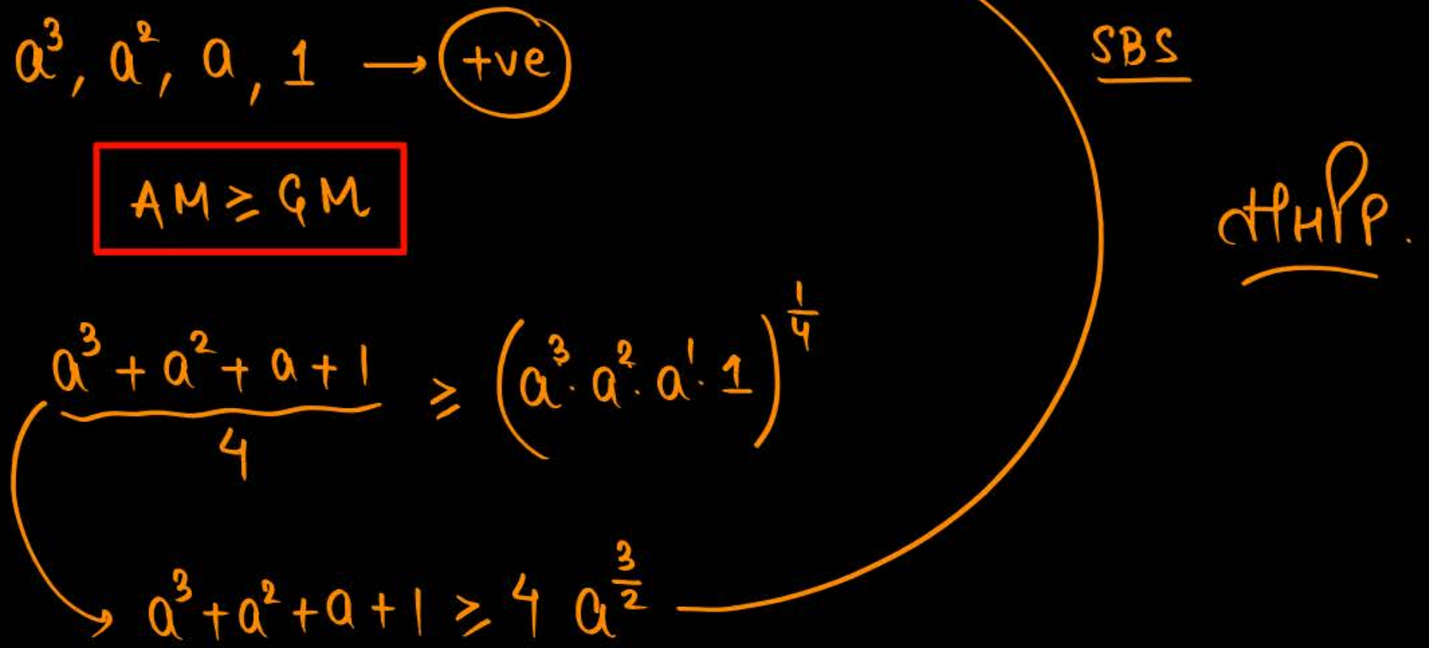
$x^2 = 1$

$x = \pm 1$

$x = 1$ ($x \neq -1$)

Question

If $a > 0$ prove that $(a^3 + a^2 + a + 1)^2 \geq 16a^3$.



DO IT BY YOURSELF (DIBY)

DIBY-64

Let a and b be positive real numbers. If a, A_1, A_2, b are in arithmetic progression, a, G_1, G_2, b are in geometric progression and a, H_1, H_2, b are in harmonic progression, show that $\frac{G_1G_2}{H_1H_2} = \frac{A_1+A_2}{H_1+H_2} = \frac{(2a+b)(a+2b)}{9ab}$.

DIBY-65

Find the sum $2017 + \frac{1}{4}\left(2016 + \frac{1}{4}\left(2015 + \cdots + \frac{1}{4}\left(2 + \frac{1}{4}(1)\right)\ldots\right)\right)$

[Ans. : $S = \frac{4}{3}(2017) - \frac{4}{9}\left(1 - \frac{1}{4^{2017}}\right)$]

DIBY-66

Find the sum $1 + 2\left(1 + \frac{1}{50}\right) + 3\left(1 + \frac{1}{50}\right)^2 + \cdots 50$ terms .

[Ans. : 2500]

DIBY-67

Find the sum $\frac{1^2}{2} + \frac{3^2}{2^2} + \frac{5^2}{2^3} + \frac{7^2}{2^4} + \cdots \infty$

[Ans. : 17]

A.G.P.

DIBY-68

Let α and β be the roots of equation $px^2 + qx + r = 0, p \neq 0$. If p, q, r, are in A.P. and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$ is: [JEE Main-2014]

(A) $\frac{\sqrt{61}}{9}$ (B) $\frac{2\sqrt{17}}{9}$ (C) $\frac{\sqrt{34}}{9}$ (D) $\frac{2\sqrt{13}}{9}$

[Ans. : (D)]

DIBY-69

If b, a, c are in Arithmetic Progression (A.P.) and c, b, a are in Geometric Progression (G.P.), then the sequence a, c, b is in.

(A) Arithmetic Progression (A.P.)
(B) Geometric Progression (G.P.)
(C) Harmonic Progression (H.P.)
(D) Arithmetico-Geometric Progression (A.G.P.)

[Ans. : (C)]

WRITING T_n FOR SERIES WITH COMMON DIFFERENCE IN AP OR GP

DIBY-84

S_n = 2 + 5 + 14 + 41 + 122 + ... Upto n-terms.

[Ans. : $\frac{3^{n+1}+2n-3}{4}$]

DIBY-85

The sum $1 + 3 + 11 + 25 + 45 + 71 + \cdots$ upto 20 terms, is equal to

[JEE Mains-2025]

(A) 7240 (B) 7130 (C) 6982 (D) 8124

[Ans. : (A)]

DIBY-86

If $S_n = 4 + 11 + 21 + 34 + 50 + \cdots$ to n terms, then $\frac{1}{60}(S_{29} - S_9)$ [JEE Mains-2023]

(A) 226 (B) 220 (C) 223 (D) 227

[Ans. : (C)]

WRITING T_n FOR SERIES WITH COMMON DIFFERENCE IN AP OR GP

DIBY-87

Find the sum of the series 6 + 13 + 22 + 33 + ... up to n terms.

(A) $\frac{n}{6}(2n^2 + 15n + 19)$

(B) $\frac{n}{6}(2n^2 + 13n + 21)$

(C) $\frac{n}{3}(n^2 + 7n + 10)$

(D) $\frac{n}{6}(2n^2 + 11n + 17)$

[Ans. : (A)]

TELESCOPIC METHOD OF DIFFERENCE

DIBY-88

$\frac{1}{3^2-1}+\frac{1}{5^2-1}+\frac{1}{7^2-1}+\cdots+\frac{1}{(201)^2-1}$ is equal to [JEE Mains-2021]

(A) $\frac{101}{404}$ (B) $\frac{25}{101}$ (C) $\frac{101}{408}$ (D) $\frac{99}{400}$

[Ans. : (B)]

DIBY-89

If the sum of series $\frac{1}{1\cdot(1+d)}+\frac{1}{(1+d)(1+2d)}+\cdots+\frac{1}{(1+9d)(1+10d)}$ is equal to 5, then $50d$ is equal to: [JEE Mains-2024]

(A) 20 (B) 5 (C) 15 (D) 10

[Ans. : (A)]

DIBY-90

The sum $\sum_{n=1}^{21}\frac{3}{(4n-1)(4n+3)}$ is equal to [JEE Mains-2022]

(A) $\frac{7}{87}$ (B) $\frac{7}{29}$ (C) $\frac{14}{87}$ (D) $\frac{21}{29}$

[Ans. : (B)]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: Only 2 Exercise

Topicwise & Bridge(Setu) - 4 sittings each of 1 HR

4HR

It is never about end result

It's always about Journey

Thank You

#futureIITians